

Relative-gauge composition for quadratic sign discrepancy

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Abstract

For a symmetric zero-diagonal sign matrix A , put $Q_A(x) = \frac{1}{2}x^\top Ax$, $M(A) = \max_x |Q_A(x)|$, and $F(n) = \min_A M(A)$. The unresolved question is whether $F(n)/n^{3/2}$ converges. We isolate a sufficient power-saving near-subadditivity theorem and derive an exact block-composition law on the group of relative switchings. The law has both a zero-temperature max-plus form and a finite-temperature conditional-alignment form. It shows precisely why separate scalar bounds on the two internal blocks and the rectangular cross block lose the leading order. Unlabeled energy profiles give a sharp order-statistic relaxation, while labeled Fourier shells retain the missing optimizer alignment. The remaining problem is a quantitative empty-fiber or good-fiber theorem at the power-saving threshold.

Status. The central identities and reduction statements are proved below; the closed-route table summarizes proofs in the accompanying archive. The labeled fiber estimate in Section 7 is explicitly conjectural. The MathOverflow limit question remains open.

1 The problem and a convergence criterion

Let

$$Q_A(x) = \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j, \quad M(A) = \max_{x \in \{-1, 1\}^n} |Q_A(x)|, \quad F(n) = \min_A M(A).$$

The audited global bounds are

$$\frac{1}{\pi} \leq \liminf_n \frac{F(n)}{n^{3/2}} \leq \limsup_n \frac{F(n)}{n^{3/2}} \leq \frac{1}{2}.$$

Neither bound compares optimizers at different orders.

The correct scalar for a composition theorem is

$$H(n) = F(n)^{2/3}.$$

Theorem 1.1 (Power-saving near-subadditivity settles convergence). *Suppose there are constants $C < \infty$ and $\delta > 0$ such that for all $n, m \geq 1$,*

$$H(n + m) \leq H(n) + H(m) + C(n + m)^{1-\delta}. \quad (1)$$

Then $H(n)/n$ converges, and therefore $F(n)/n^{3/2}$ converges.

Proof. Replace δ by $\min(\delta, 1)$. The known upper bound gives $H(n) = O(n)$. Fix a block length k and combine q copies along a balanced binary tree. At the level where a subtree has at most 2^j leaves, there are at most $\lceil q/2^j \rceil$ merges, each with defect $O((2^j k)^{1-\delta})$. Summing levels gives

$$H(qk) \leq qH(k) + O(qk^{1-\delta}),$$

uniformly in q ; the geometric factor is $\sum_{j \geq 1} 2^{-j\delta} < \infty$, while the incomplete-tree remainder is no larger.

For $N = qk + r$, $0 \leq r < k$, one final merge and $H(r) = O(k)$ give

$$\limsup_{N \rightarrow \infty} \frac{H(N)}{N} \leq \frac{H(k)}{k} + O(k^{-\delta}).$$

Choose $k \rightarrow \infty$ along a sequence on which $H(k)/k$ approaches its liminf. The limsup is then no larger than the liminf. Finally $F(n)/n^{3/2} = (H(n)/n)^{3/2}$. $\square$

Using $F(t) = \Theta(t^{3/2})$, a sufficient equivalent-scale target is

$$F(n+m) \leq \left(F(n)^{2/3} + F(m)^{2/3} \right)^{3/2} + O((n+m)^{3/2-\delta}). \quad (2)$$

The power saving, or another dyadically summable modulus, is essential.

2 Cut discrepancy is equivalent but not compositional

Put $e_n = \binom{n}{2}$. For a graph G on $[n]$, define

$$\text{cdisc}_{1/2}(G) = \max_{S \subseteq [n]} \left| e_G(S, S^c) - \frac{1}{2} |S| |S^c| \right|$$

and let

$$D(n) = \min \left\{ \text{cdisc}_{1/2}(G) : e(G) = \left\lfloor \frac{e_n}{2} \right\rfloor \right\}.$$

Theorem 2.1 (Fixed-density equivalence). *For every $n \geq 2$,*

$$-1 \leq 4D(n) - F(n) \leq \sqrt{\binom{n}{2}} + 2. \quad (3)$$

Thus the two normalized sequences converge together and have the same limit.

Proof. For a signing A , let G_A contain its negative edges and let $t = \sum_{i < j} a_{ij}$. If x_S is positive on S and negative on its complement and $r = |S|(n - |S|)$, then

$$Q_A(x_S) = t - 2r + 4e_{G_A}(S, S^c).$$

On the fixed-density layer $t = t_0 \in \{0, 1\}$, this gives $|4 \text{cdisc}_{1/2}(G_A) - M(A)| \leq 1$.

For an optimal A , a uniform switch z replaces t by $Q_A(z)$. Walsh orthogonality gives $\mathbb{E}Q_A(z) = 0$ and $\mathbb{E}Q_A(z)^2 = e_n$, so some switch has $|t| \leq \sqrt{e_n}$. Flipping $|t - t_0|/2$ edges reaches the fixed-density layer and increases every absolute energy by at most $|t - t_0|$. Combining the two estimates proves (3). $\square$

This reformulation identifies the problem with dense cut discrepancy at the right scale, but ordinary graph products and dense graph limits do not retain the $n^{3/2}$ optimizer correlations needed in (2).

3 Exact max-plus composition

Fix internal signings A and B of orders n and k , a rectangular signing $C \in \{-1, 1\}^{n \times k}$, and $N = n + k$. Write

$$\mathbb{P}_r = \{-1, 1\}^r / \{\pm \mathbf{1}\}, \quad G_{n,k} = \mathbb{P}_n \times \mathbb{P}_k \times \{-1, 1\}, \quad |G_{n,k}| = 2^{N-1}.$$

Define augmented graph state spaces

$$\Omega_A = \{-1, 1\} \times \mathbb{P}_n, \quad e_A(\sigma, [x]) = \sigma Q_A(x),$$

and analogously Ω_B, e_B . Let

$$\overline{\mathcal{R}}_{n,k} = \{[xy^\top] : x \in \{-1, 1\}^n, y \in \{-1, 1\}^k\}, \quad e_C([R]) = |\langle C, R \rangle|,$$

where R and $-R$ are identified.

There is a balanced homomorphism

$$\pi((\sigma, [z]), (\eta, [w]), [xy^\top]) = ([zx], [wy], \sigma\eta) \quad (4)$$

onto $G_{n,k}$. Its domain has 2^{2N-2} elements, hence every fiber has 2^{N-1} elements.

Put

$$M_A = M(A), \quad M_B = M(B), \quad B_C = \max_{x,y} |x^\top C y|, \quad L = M_A + M_B + B_C,$$

and define deficits

$$d_A = M_A - e_A, \quad d_B = M_B - e_B, \quad d_C = B_C - e_C.$$

For $g = (\alpha, \beta, \tau) \in G_{n,k}$, set

$$Y_g = \begin{pmatrix} D_\alpha A D_\alpha & C \\ C^\top & \tau D_\beta B D_\beta \end{pmatrix}.$$

Theorem 3.1 (Relative-gauge max-plus identity). *For every $g \in G_{n,k}$,*

$$L - M(Y_g) = \min_{\pi(a,b,r)=g} (d_A(a) + d_B(b) + d_C(r)). \quad (5)$$

If all 2^{2N-2} product-triple deficit sums are listed with multiplicity as $\lambda_1 \leq \dots \leq \lambda_{2^{2N-2}}$ and $\Lambda = \lambda_{2^{N-1}}$, then

$$F(N) \leq \min_g M(Y_g) \leq L - \Lambda. \quad (6)$$

Equivalently, if fewer than 2^{N-1} triples have total deficit strictly below s , some gauge has $M(Y_g) \leq L - s$.

Proof. Choose representatives $a = (\sigma, [z])$, $b = (\eta, [w])$, and $r = [xy^\top]$, orienting $R = xy^\top$ so that its cross energy has sign σ . Evaluating $Y_{\pi(a,b,r)}$ at (x, y) and multiplying by the appropriate global sign gives $e_A(a) + e_B(b) + e_C(r)$. Conversely, every augmented full spin determines such a triple; replacing its signed cross value by its absolute value can only increase the local sum. Therefore

$$M(Y_g) = \max_{\pi(a,b,r)=g} (e_A(a) + e_B(b) + e_C(r)),$$

which is (5) after subtracting from L .

Choose one minimizing triple from each of the 2^{N-1} fibers. They are distinct, so the largest selected minimum is at least the 2^{N-1} -st global order statistic. This proves (6). $\square$

The order statistic is optimal if all additive labels are discarded: one can place the 2^{N-1} smallest deficits one per equal fiber. It is therefore a sharp scalar relaxation, not the complete composition state.

4 Labeled occupancy and Fourier factorization

For a threshold s , define the exact fiber occupancy

$$b_s(g) = \#\{(a, b, r) : \pi(a, b, r) = g, d_A(a) + d_B(b) + d_C(r) < s\}. \quad (7)$$

By Theorem 3.1, $b_s(g) = 0$ if and only if $M(Y_g) \leq L - s$.

Characters of $G_{n,k}$ are indexed by even sets $I \subseteq [n]$, $J \subseteq [k]$, and $\epsilon \in \{0, 1\}$. For a deficit value d , put

$$\begin{aligned} A_d(I, \epsilon) &= \sum_{d_A(\sigma, [z])=d} \sigma^\epsilon z_I, \\ B_d(J, \epsilon) &= \sum_{d_B(\eta, [w])=d} \eta^\epsilon w_J, \\ C_d(I, J) &= \sum_{d_C([xy^T])=d} x_I y_J. \end{aligned}$$

Proposition 4.1 (Exact labeled-shell factorization). *Let $K = 2^{N-1}$ and normalize Fourier coefficients by $\widehat{b}(\chi) = K^{-1} \sum_g b(g) \chi(g)$. Then*

$$\widehat{b}_s(I, J, \epsilon) = \frac{1}{K} \sum_{d_A+d_B+d_C < s} A_{d_A}(I, \epsilon) B_{d_B}(J, \epsilon) C_{d_C}(I, J). \quad (8)$$

If $\mu_s = K^{-1} \sum_g b_s(g)$ and $V_s = \sum_{\chi \neq 1} |\widehat{b}_s(\chi)|^2$, then

$$\min_g b_s(g) \leq \mu_s - \sqrt{\frac{V_s}{K-1}}. \quad (9)$$

In particular, a right side smaller than one certifies an empty fiber.

Proof. The character (I, J, ϵ) pulls back under (4) to $\sigma^\epsilon z_I \eta^\epsilon w_J x_I y_J$. Summing independently over the three labeled shells proves (8).

For (9), put $h = b_s - \mu_s$ and $m = -\min h$. Since $h \geq -m$ and $\sum_g h(g) = 0$, also $h \leq (K-1)m$. Averaging $(h+m)(h-(K-1)m) \leq 0$ gives $V_s = K^{-1} \sum_g h(g)^2 \leq (K-1)m^2$. $\square$

This proposition already improves the scalar order statistic in exact small examples. At the balanced order-fourteen Paley split and the threshold corresponding to energy twenty-seven, however,

$$K = 8192, \text{quad} \sum_g b_s(g) = 304908, \text{quad} \min_g b_s(g) = 0, \text{quad} \max_g b_s(g) = 87.$$

There is exactly one empty fiber, while the variance bound remains above one. The good gauge is real but exponentially isolated. Generic moment or polynomial-precision methods cannot reliably detect this signal without using the exact shell labels; the repository contains explicit matching-moment and Fourier-support obstructions.

5 Finite-temperature alignment

The same state appears at finite temperature. Define the augmented partition function and its annealed normalization by

$$Z_A(t) = \sum_{x \in \{-1,1\}^n, \sigma \in \{-1,1\}} e^{t\sigma Q_A(x)}, \quad f_{n,t}(A) = \frac{Z_A(t)}{2^{n+1}(\cosh t)^{\binom{n}{2}}},$$

and

$$\mathcal{G}_n(q, t) = \log \mathbb{E}_A f_{n,t}(A)^{-q}.$$

For the rectangular block use

$$Z_C^\square(t) = \sum_{x,y} e^{tx^\top Cy}, \quad f_{n,k,t}^\square(C) = \frac{Z_C^\square(t)}{2^N(\cosh t)^{nk}},$$

with reverse moment $\mathcal{G}_{n,k}^\square$.

Theorem 5.1 (Conditional-alignment chain). *For $n, k \geq 3$,*

$$\mathcal{G}_N(q, t) = \mathcal{G}_n(q, t) + \mathcal{G}_k(q, t) + \mathcal{G}_{n,k}^\square(q, t) + \mathcal{A}_{n,k}(q, t), \quad (10)$$

where $\mathcal{A}_{n,k}(q, t) \geq 0$ is the reverse moment of the conditional density on the 2^{N-1} relative-switching states.

Proof. The global cut code is a subcode of the direct sum of the two internal cut codes and the rectangular row-column switching code. Their quotient fiber has size 2^{N-1} . A coarse state (A, B, C) has fine representatives

$$Y_{\alpha,\beta,\tau} = \begin{pmatrix} D_\alpha A D_\alpha & C \\ C^\top & \tau D_\beta B D_\beta \end{pmatrix}.$$

Direct spin changes give

$$\mathbb{E}_{\alpha,\beta,\tau} Z_Y(t) = \frac{Z_A(t) Z_B(t) Z_C^\square(t)}{2^{N+1}}.$$

Thus the conditional density relative to the uniform fiber is

$$r_{A,B,C}(\alpha, \beta, \tau) = \frac{2^{N+1} Z_Y(t)}{Z_A(t) Z_B(t) Z_C^\square(t)}, \quad \mathbb{E}_{\text{fiber}} r = 1.$$

Grouping the fine negative moment first by coarse states gives (10); convexity of $z \mapsto z^{-q}$ gives the nonnegativity of the logarithmic correction. $\square$

The correction is leading order. Let $\mathcal{A}_{n,k}^\infty(t)$ denote its $q \rightarrow \infty$ slope, and put

$$B_\square(n, k) = \min_C \max_{x,y} |x^\top Cy|.$$

Finite-space Laplace limits give

$$\lim_{t \rightarrow \infty} \frac{\mathcal{A}_{n,k}^\infty(t)}{t} = F(n) + F(k) + B_\square(n, k) - F(n+k). \quad (11)$$

Moreover

$$\liminf_n \frac{2F(n) + B_\square(n, n) - F(2n)}{n^{3/2}} \geq \frac{2}{\pi} + \sqrt{\frac{2}{\pi}} - \sqrt{2} > 0.$$

Therefore negative replicas do not bypass optimizer-compatible composition; they recover its exact finite-temperature state.

The first nonconstant term of the conditional log-density is already mixed. Writing $P = D_\alpha A D_\alpha$ and $Q = D_\beta B D_\beta$, its order-four Eulerian Hamiltonian is

$$\mathcal{H}_4 = \frac{1}{2} \left[\text{tr}(P^2 C C^\top) + \text{tr}(Q^2 C^\top C) + \tau \text{tr}(P C Q C^\top) - nk(N-2) \right]. \quad (12)$$

This depends on relative eigenvector and row-correlation geometry, not merely on the spectra or scalar partition functions of the three blocks.

6 Closed shortcuts

The following conclusions concern specific mechanisms, not the original problem.

Mechanism	Audited obstruction
Cube relaxation	The unique optimizer is the zero coupling; the sign integrality gap is $\Theta(n^{3/2})$.
Hamming union bound	Its balanced certificate has normalized floor $\sqrt{\log 2/2} > 1/2$.
Scalar negative-replica transport	Reverse hypercontractivity follows the correct parameter curve but its defect is $\Omega(n^2)$ in a rigorous parameter range.
Graph plus rectangular transport	The rectangular marginal cannot pay that defect in the zero-temperature regime; the alignment term is leading.
Scalar energy profiles	They give the sharp unlabeled order statistic but finite collisions and the order-fourteen calibration show that additive labels can carry the required gain.
Low moments and generic vacancy tests	An isolated empty fiber can be hidden to exponentially high moment degree in the abstract occupancy problem.
Graphons and empirical spectra	Their dense normalizations erase the $n^{3/2}$ fluctuation scale.

This list is a reason to narrow the state, not a claim that every analytic or probabilistic approach is impossible.

7 The settling target

For fixed A, B, C , define the desired composition ceiling

$$T_{n,k} = \left(F(n)^{2/3} + F(k)^{2/3} \right)^{3/2} + C_0 N^{3/2-\delta}$$

and the required deficit $s = L - T_{n,k}$. Theorem 3.1 turns the entire asymptotic problem into the following precise target.

Definition 7.1 (Labeled good-fiber target). There exist $C_0 < \infty$, $\delta > 0$, and a rule assigning to every pair n, k internal signings A, B and a rectangular signing C such that

$$\min_{g \in G_{n,k}} b_{L-T_{n,k}}(g) = 0. \quad (13)$$

An abundance version requires at least 2^{cN} gauges satisfying the same inequality for some $c > 0$ over proportional splits.

Proposition 7.2 (Why this target settles the question). *If (13) holds uniformly for all n, k , then (2) holds and $F(n)/n^{3/2}$ converges. The same conclusion follows from any abundance theorem with the stated energy threshold.*

Proof. An empty fiber gives $M(Y_g) \leq L - (L - T_{n,k}) = T_{n,k}$ by Theorem 3.1. Hence $F(N) \leq T_{n,k}$. This is (2), so Theorem 1.1 applies. $\square$

The proposition is a reduction, not a proof of the missing estimate. Its value is that every future invariant can be tested against one exact object: the labeled occupancy b_s and its factorization (8). The strongest immediate research targets are:

1. a character-sum sign theorem forcing one empty fiber at the required threshold;
2. a large-deviation theorem producing exponentially many good gauges;
3. a variational limit for the full relative-switching law with a power-saving, dyadically summable composition defect.

Further exact values of $F(n)$ are useful only when they identify or falsify a candidate labeled state. The companion finite-results manuscript records the current exact table and finite structural theorems; the broad research archive preserves all other routes and counterexamples.

References

- [1] P. Ivanisvili, *Min-max of a quadratic form of plus-minus ones*, MathOverflow question 413935, 2022.
- [2] C. Borell, *Positivity improving operators and hypercontractivity*, Mathematische Zeitschrift 180 (1982), 225–234.
- [3] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, Journal of Symbolic Computation 60 (2014), 94–112.